MARWAN TAREK

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SUMMARY

Software Engineer with 3+ years in **embedded software testing and validation** for safety-critical systems: automotive **ADAS ECUs** at **Valeo** and railway signalling at **Alstom**. Hands-on with **SiL/HiL** testing, **Vector CANoe/CANalyzer**, **vTESTstudio**, **Python** and **Robot Framework** test automation, **requirements-based testing** and **traceability** (**DOORS**), **CI/CD** quality gates, and **C/C++** on **RTOS**, working to the **V-Model**, **ISO 26262**, **ASPICE** and **MISRA C**.

EXPERIENCE

Alstom — Software Engineer Jul 2025 – Present *Cairo, Egypt*

- Automated **requirements-based testing**: generated test cases from system requirement specifications and measured **requirement coverage** and **traceability** across system and product levels.
- Added **Robot Framework** suites and quality gates to **GitLab CI** and **Jenkins**, raising regression coverage by about **30%**.
- Built a **C++** validation and diagnostics tool that reduced manual bench time, and automated test flows in **Python**.
- Built a **C#/.NET** service that decodes, validates and routes **Protobuf** alarm and train-status messages over **AMQP** (ActiveMQ Artemis), with a **playback mode** for deterministic tests.

Spark Systems — Software Engineer (.NET) Jan 2025 – Jun 2025 *Giza, Egypt*

- Built and tested **C#/ASP.NET Core** REST APIs; validated workflows with **Postman** and CI checks in **Jenkins**.

Valeo — Embedded Software Engineer Jul 2022 – Oct 2023 *Smart Village, Egypt*

- Ran **SiL/HiL** tests for **ADAS** ECUs (LiDAR/camera) with **Vector CANoe/CANalyzer**, **winIDEA (iSYSTEM)**, **JTAG/SWD** and oscilloscopes, measuring execution time and CPU load.
- Wrote and executed functional and non-functional tests in the **V-Model** with **vTESTstudio** and **Python**, integrated into **Jenkins**; traced tests to **IBM DOORS** requirements and reported defects in **Jira**.
- Validated **bootloader**/**UDS** flashing and **CAN**/**SPI**/**I2C**/**UART**/Ethernet drivers on an **RTOS**; developed multithreaded **C/C++** on **AUTOSAR**.
- Followed **ISO 26262**, **ASPICE** and **MISRA C** with **static analysis**, **code coverage** and requirements traceability.

PROJECTS (GITHUB)

- **UDS Diagnostics Client** (C++17, CMake): diagnostics library and CLI with a pluggable transport layer (mock, Linux **SocketCAN** ISO-TP), JSON/CSV reports and **CTest** unit tests.
- **Dual-Bank CAN Bootloader** (C, STM32F4): A/B boot over **UDS/ISO-TP** with CRC32 and rollback, unit-tested on a host flash simulator; **Python** flasher.
- **UDS Test Bench** (Robot Framework): UDS test suites in CI; **MISRA Pack** (Cppcheck): static analysis in CI; **E2E Test Lab** (Robot, Playwright, k6): tests on every push.

SKILLS

Testing: SiL/HiL, V-Model, requirements-based, integration and system testing, regression, traceability
Tools: Vector CANoe/CANalyzer, vTESTstudio, winIDEA, JTAG/SWD, oscilloscope, IBM DOORS, Jira
Automation: Python, PyTest, Robot Framework, CTest, Playwright, Jenkins, GitLab CI, Docker, Git, Linux
Embedded: C, C++17, RTOS/FreeRTOS, STM32 (Cortex-M), AUTOSAR, CAN, UDS, SPI, I2C, UART
Standards: ISO 26262, ASPICE, MISRA C, static analysis, code coverage, Agile/Scrum

EDUCATION

B.Sc. in Computer Science (Honours), Helwan University Sep 2019 – Jul 2023
Languages: English (C2), French (A2), Arabic (Native)